BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER VI - END SEMESTER EXAMINATION

December 2023

Paper Code: BCA-604

Subject: Operating Systems

Time Allowed: 3 Hours | Maximum Marks: 75

General Instructions:

1. Attempt all sections. Section A is compulsory; internal choice is given in Sections B and C.
2. Write answers in legible handwriting and number them exactly as in the question paper.
3. Draw neat, labelled diagrams wherever required.

SECTION A (5 x 3 = 15 Marks)

Answer all five questions. Each question carries 3 marks.

1. Define a process. [3]

2. What is thrashing? [3]

3. Differentiate between paging and segmentation. [3]

4. What is a deadlock? [3]

5. What is a system call? [3]

SECTION B (4 x 10 = 40 Marks)

Answer any four questions. Each question carries 10 marks.

1. Explain the process state diagram with all transitions. [10]

2. Compare FCFS, SJF and Round Robin CPU scheduling with examples. [10]

3. Explain deadlock prevention and deadlock avoidance. [10]

4. Describe virtual memory and demand paging. [10]

SECTION C (2 x 10 = 20 Marks)

Answer any two questions. Each question carries 10 marks.

1. Explain the critical section problem and Peterson's solution. [10]

2. Discuss file system structure and the FCFS, SSTF and SCAN disk scheduling algorithms. [10]

*** *END OF PAPER* ***